Query-Guided Asymmetric Grid Composition: A Training-Free Approach for Long Video Understanding

Group 9

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Abstract

Vision–Language Models (VLMs) perform well on short video question answering, but their performance degrades substantially as video duration increases. The core challenge is a fixed visual-token budget: long videos must be compressed either by sampling fewer frames, which risks losing temporal coverage, or by reducing per-frame resolution, which can remove fine-grained spatial details required for accurate reasoning. Existing training-free approaches typically address only one side of this trade-off. Query-aware methods such as BOLT and Q-Frame prioritize frames relevant to the question but present them as separate images, while Video Panels improves temporal coverage by composing multiple frames into fixed, equal-sized grids regardless of frame relevance. We propose a training-free, model-agnostic approach that combines query-guided selection with adaptive visual composition. We densely sample frames, score their relevance to the question using CLIP, select a temporally distributed subset, and compose the selected frames into asymmetric 2×2 panels whose cell areas are determined by relevance scores. This allocation preserves more pixels for highly relevant frames, compresses less relevant frames rather than discarding them, and maintains chronological reading order. Using Qwen2.5-VL-7B-Instruct on the medium and long splits of Video-MME, our method achieves 58.7% accuracy compared with 57.8% for the paneling baseline, with gains concentrated in spatial, temporal, and OCR questions and regressions in recognition and summary tasks. On MF2, a long-movie true/false benchmark, pair accuracy improves from 16.9% to 23.5% ($p = 0.0016$), accompanied by a reduced bias toward predicting false. Overall, our method raises the bar for long video understanding models.

1 Introduction

Long Video Understanding (LVU) is the task of answering natural-language questions by reasoning over videos that may run for minutes or hours. It is harder than short-video understanding for reasons that compound each other. A one-hour video at 24 frames per second contains close to 90,000 frames, far beyond what any current model can process [2]. Most of those frames are near-duplicates of their neighbours, so processing them all would waste computation. The moments that actually answer a question are rare and short, and the evidence needed for one answer is often spread across events that are far apart in time, which means the model must combine several pieces of evidence rather than look at a single frame [9, 21].

Multimodal Large Language Models have made rapid progress on image understanding [11, 8], and that progress has naturally been extended to video. Performance, however, degrades sharply with video length. Qwen2.5-VL shows a marked drop in accuracy on videos longer than three

minutes [20], and the cause is architectural rather than perceptual: the context windows of the underlying language models are small relative to what a long video contains. VideoLLaMA2 supports roughly 2,000 visual tokens and VILA-V1.5 about 4,000 [15, 16], so feeding every frame of a long video is simply not possible. A great deal of recent work tries to relax this bound from inside the model, by compressing or pruning visual tokens before temporal modelling [13, 22], by extending the language backbone's context length [14], or by adding memory and summarisation modules [23]. These methods add parameters, training, or both, and many of them still struggle to consistently beat the base model they are built on [3].

Because full coverage is impossible, every practical system applies a sampling function that reduces the video to T frames, and in nearly all cases that function is uniform sampling. Uniform sampling is simple and unbiased, but it has three well-documented drawbacks [2]. It is *sparse*, so as videos get longer the sampled frames drift further apart and short transitions fall between them. It is *static and query-agnostic*, so the same frames are shown to the model no matter what is being asked. And it *treats all frames equally*, even though the information a question depends on is concentrated in a handful of moments. Query-conditioned selectors were introduced to address the second point, through text–frame matching [17], learned frame ranking [18], inverse transform sampling [1], or CLIP scoring with resolution assignment [2].

Underneath all of this sits what we call the **spatio-temporal resolution mismatch**. A VLM has a fixed context window, and therefore a fixed budget of visual tokens. Given a long video, that budget can be spent in two ways. Spend it on a few frames at full resolution, and the model sees sharp images but misses most of the video's timeline. Spend it on many frames at reduced resolution, and the model covers the timeline but can no longer read a licence plate, a scoreboard or a caption. VLMs are trained mostly on images and short clips, where both resolutions stay high after sampling; for long videos the temporal resolution collapses while the spatial resolution does not, so a disproportionate share of the model's capacity is spent on spatial detail that the question may not need [3]. Existing methods effectively choose one of these two positions in advance, for every question, which is the limitation our work targets.

Two recent training-free approaches sit on opposite sides of this choice, and together they motivate our design. Q-Frame [2] selects query-relevant frames with CLIP [7] and assigns each one of three resolutions according to its rank, but passes the selected frames to the model as a list of separate images, so nothing in the input encodes how they are ordered or how far apart they are in time. Video Panels [3] instead stitches several frames into one composite image, like panels on a comic page, which raises temporal coverage fourfold within the same token budget and improves eight VLMs across five benchmarks. Its grid, however, is rigid and query-blind: every frame is shrunk by the same factor whether or not it contains the answer. At the other end of the spectrum, agentic systems such as Symphony [4] achieve both goals by decomposing the task across specialised agents, at the cost of multi-round inference for every question.

We take the view that this choice should not be made in advance and should not be made uniformly. Within a single video, some frames matter a great deal for a given question and others are context. A system that knows which is which can spend its pixel budget unevenly: give the query-relevant frames a large share of the canvas, keep the rest at a small size so that temporal continuity survives, and never discard a frame that still carries context. This is the idea we develop and evaluate in this report. Our method scores densely sampled candidate frames against the question with CLIP, keeps a temporally spread subset, and composes the survivors into 2×2 panels whose cell areas are proportional to relevance. It is training-free, model-agnostic, and runs in a single forward pass.

Contributions. The work presented here makes the following contributions.

1. A training-free preprocessing pipeline that combines query-aware frame selection with composite paneling, so that a VLM receives both dense temporal coverage and high spatial resolution where it matters.

2. An *asymmetric* 2×2 panel layout in which each cell's area is a function of its CLIP relevance score. This gives every frame a resolution that reflects its importance to the query, instead of applying one fixed resolution to all of them.
3. A design for embedding temporal sequence information directly into the composed panels, so that the ordering and spacing of frames is made explicit to the VLM rather than left to be inferred.

2 Literature Review and Background Studies

2.1 Video-Language Models and the context bottleneck

Modern VLMs such as Qwen2.5-VL [8], LLaVA-OneVision [11] and LLaVA-Video [12] treat a video as a sequence of images, encode each image into visual tokens, and feed those tokens to a language model together with the question. Qwen2.5-VL, the backbone we use, converts an image into roughly one token per 28×28 pixel block after an internal resize, so a 1024×576 image costs about 777 visual tokens. The number of frames a model can accept is therefore bounded by its context window, and a large body of work tries to relax that bound, for instance by compressing tokens, by extending the context length, or by adding memory modules [13, 14]. These approaches change the model. Our interest is in the complementary direction: keeping the model fixed and changing what we show it.

2.2 Query-aware frame selection

BOLT [1] improves long-video question answering at inference time by using inverse transform sampling to prioritise query-relevant frames, balancing relevance against temporal diversity, with no retraining. Q-Frame [2] scores uniformly sampled candidate frames against the query with CLIP, uses the Gumbel-Max trick to sample a diverse top-K subset, and then assigns each selected frame one of three resolutions according to its rank, so that important frames keep detail and background frames are compressed. Both report solid gains and both are training-free, which is the property we want to preserve.

Their shared limitation is structural. Selected frames are passed to the model as a list of separate images. Nothing in the input tells the model that two images are adjacent in time or how far apart they are, so the temporal narrative has to be reconstructed by the language model from visual cues alone. Q-Frame's own authors note this in their failure analysis: on temporal reasoning questions, individually relevant frames often fail to capture the transitions that the question is actually about.

2.3 Visual prompting and composite images

Video Panels [3] takes a different route. Rather than selecting better frames, it changes the format of the input: $\alpha\beta$ consecutive frames are downscaled and stitched into a single composite image, like panels on a comic page, and a sequence of such composites is given to the model. With $\alpha = \beta = 2$, this multiplies the number of frames the model sees by four within the same token budget. The method is training-free and model-agnostic, and it improves accuracy across eight VLMs and five benchmarks, with the largest gains on the longest videos. A closely related idea appears in work on "super images" for video retrieval [5] and in image-grid approaches to short-video question answering [6].

The limitation here is symmetry. Video Panels samples uniformly, so it is query-blind, and its grid is rigid: every frame is shrunk by exactly the same factor whether or not it contains the answer. For a question that hinges on small text or a small object, the relevant frame is compressed as aggressively as the thirty-one frames that do not matter.

2.4 Agentic approaches

Systems such as Symphony [4] decompose long-video tasks across specialised agents for planning, grounding, perception and subtitle reading, coordinated through multi-round collaboration with reflection. These systems are accurate but expensive: each question triggers several model calls. Our aim is to obtain some of the same benefit within a single forward pass.

2.5 Research gap

Table 1 summarises the gap our work addresses. Each existing family solves one part of the problem and leaves another open. Query-aware selection finds the right frames but presents them without temporal structure. Paneling supplies temporal structure but allocates space blindly. Agentic reasoning achieves both at a computational cost that rules out large-scale evaluation.

Table 1: The research gap addressed in this work.

Existing strategy	**Gap**	**What is needed**
Visual consolidation (paneling)	Fixed spatial allocation	Adaptive, query-driven space
Query-aware frame selection	Temporal discontinuity	Chronological composition
Multi-agent reasoning	Heavy computational cost	Single-pass, training-free inference

3 Methodology

3.1 Problem formulation

A video is a sequence of frames $V = \{f_1, \ldots, f_D\}$ and the input to the system is a pair (V, q), where q is a multiple-choice question together with its answer options. The VLM can accept at most C images, so the task of a preprocessing method is to construct a set of C images $\mathcal{P} = \{P_1, \ldots, P_C\}$ from V such that the model's answer $a = \mathrm{VLM}(\mathcal{P}, q)$ is correct as often as possible. Video Panels constructs $\mathcal{P}$ by uniform sampling and equal-size tiling. We construct it in four stages, illustrated in Figure 1 and described below.

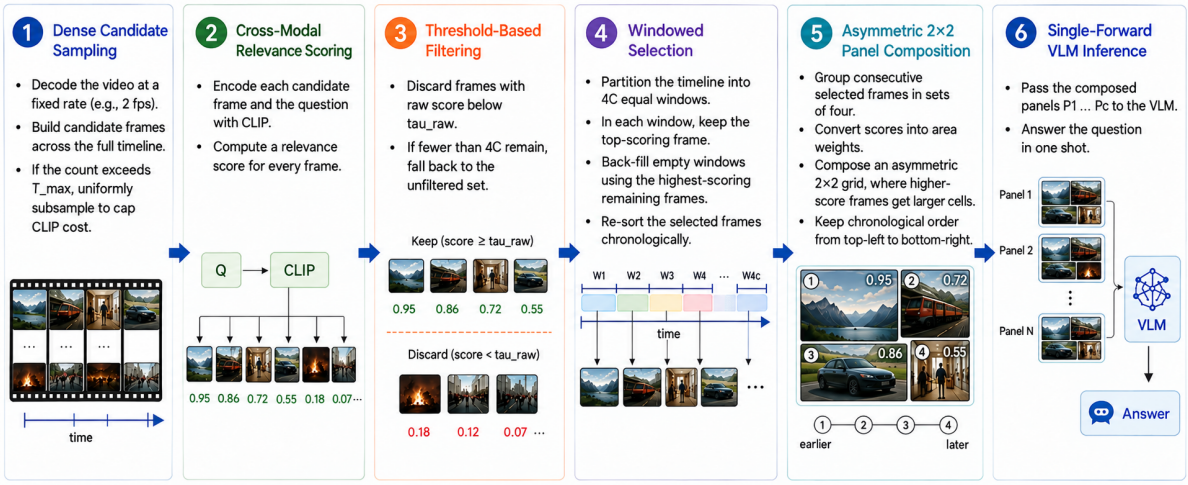


Figure 1: Methodology Pipeline

3.2 Stage 1: dense candidate sampling

We decode the video at a fixed rate ρ rather than taking a fixed number of frames, so that longer videos contribute more candidates:

$$\mathcal{F} = \{(f_j, t_j)\}_{j=1}^{T}, \qquad t_j = \frac{j \cdot \lfloor \mathrm{fps}/\rho \rceil}{\mathrm{fps}}, \qquad T \le T_{\max}. \tag{1}$$

Candidates are decoded at 640×360 to bound memory, and if the sampling rate yields more than $T_{\max}$ frames we subsample the index list uniformly, so that the CLIP cost per video is capped regardless of duration. We use $\rho = 2$ fps with $T_{\max} = 1024$ for the Video-MME experiments and a lower rate with $T_{\max} = 2048$ for MF2, where videos are full-length movies.

3.3 Stage 2: cross-modal relevance scoring

Each candidate frame and the question are embedded with a frozen CLIP model [7] (ViT-B/32) into a shared space, and relevance is the cosine similarity between the two:

$$\mathbf{u}_j = \frac{E_{\mathrm{img}}(f_j)}{\|E_{\mathrm{img}}(f_j)\|}, \quad \mathbf{v} = \frac{E_{\mathrm{txt}}(q)}{\|E_{\mathrm{txt}}(q)\|}, \qquad s_j = \mathbf{u}_j^{\top}\mathbf{v}. \tag{2}$$

Frames whose raw similarity falls below an absolute floor τ_{raw} are discarded:

$$\mathcal{F}' = \{(f_j, t_j, s_j) \mid s_j \ge \tau_{\mathrm{raw}}\}. \tag{3}$$

We set $\tau_{\mathrm{raw}} = 0.15$. The threshold is deliberately applied to the raw similarity rather than to min–max normalised scores: raw CLIP text–image similarities occupy a narrow band of roughly 0.15 to 0.35, so an absolute floor means "this frame is genuinely unrelated to the question", whereas a normalised threshold would mean "this frame is among the weaker frames of this particular video", which is a different and less useful statement. If filtering leaves fewer than $4C$ frames, we fall back to the unfiltered set so that the frame budget is always filled.

3.4 Stage 3: windowed selection

We must choose exactly $4C$ frames, since every composite panel holds four. A greedy top-$4C$ selection is the obvious choice but fails badly on long videos: a single high-scoring scene can absorb the entire budget and the rest of the video disappears. We therefore partition the timeline into $4C$ equal windows and keep the best frame in each,

$$W_m = \Big[t_{\min} + \tfrac{(m-1)\Delta}{4C},\ t_{\min} + \tfrac{m\Delta}{4C}\Big), \qquad \hat{f}_m = \arg\max_{(f,t,s)\in\mathcal{F}'\cap W_m} s, \tag{4}$$

where $\Delta = t_{\max} - t_{\min}$. Windows that are empty after filtering are back-filled with the highest-scoring frames not yet chosen, so exactly $4C$ frames are always returned. The selected frames are then re-sorted chronologically. This selection is what distinguishes our method from a pure top-k selector such as the ablation variant we also implement: it guarantees temporal coverage while still being query-aware inside each window.

3.5 Stage 4: asymmetric composition

Consecutive groups of four selected frames are composed into one canvas each. Within a group, raw scores are converted into area weights with a sharpened softmax,

$$w_i = \frac{\exp\big((s_i - \max_k s_k)/\gamma\big)}{\sum_{k=1}^{4}\exp\big((s_k - \max_k s_k)/\gamma\big)}, \qquad \sum_{i=1}^{4} w_i = 1. \tag{5}$$

The temperature γ controls how strongly relevance translates into size. Because raw CLIP similarities within a group typically differ by only a few hundredths, a plain normalisation would give four weights all close to 0.25 and the layout would be visually indistinguishable from the symmetric baseline; we use $\gamma = 0.02$, which turns a score gap of 0.05 into a clear size difference.

A 2×2 cross-split canvas has exactly two degrees of freedom: one vertical split α and one horizontal split β. The four cell areas are therefore forced to be separable,

$$A_{\mathrm{TL}} = \alpha\beta, \quad A_{\mathrm{TR}} = (1-\alpha)\beta, \quad A_{\mathrm{BL}} = \alpha(1-\beta), \quad A_{\mathrm{BR}} = (1-\alpha)(1-\beta), \tag{6}$$

and the best available choice is to match the marginals of the target weights,

$$\alpha = \mathrm{clip}\big(w_{\mathrm{TL}} + w_{\mathrm{BL}},\, \lambda,\, 1-\lambda\big), \qquad \beta = \mathrm{clip}\big(w_{\mathrm{TL}} + w_{\mathrm{TR}},\, \lambda,\, 1-\lambda\big), \tag{7}$$

with $\lambda = 0.25$ so that no cell degenerates into an unreadable sliver (the largest cell is at most nine times the area of the smallest). We record the realised error $\epsilon = \max_i |A_i - w_i|$ for every canvas.

This construction is exact when the target weights are separable. For example, $w = [0.45, 0.25, 0.20, 0.10]$ gives $\epsilon = 0.005$. It degrades when the weights are anti-diagonal: $w = [0.40, 0.10, 0.10, 0.40]$ collapses to a uniform grid with $\epsilon = 0.15$, because no pair of straight splits can make the top-left and bottom-right cells large while keeping the other two small. This is a property of the cross-split geometry rather than a defect of the implementation. Escaping it requires a treemap-style layout, which would break the fixed top-left to bottom-right reading order; we discuss this in Section 4.10.

Each canvas is 1024×576 pixels with a 3-pixel separator so that the model can see four distinct frames rather than one collage, and the four frames are placed in chronological order: top-left is earliest, bottom-right is latest. The panels themselves are also ordered chronologically, so the sequence of canvases preserves the video's timeline.

3.6 Inference

The C canvases are passed to the VLM in a single forward pass, followed by the question, its options, and the standard Video-MME instruction to answer with the option letter only. No part of the pipeline is trained, and nothing about the model is modified, so the method can be attached to any VLM that accepts multiple images.

3.7 Implementation

The pipeline is implemented in Python with OpenCV for decoding, Hugging Face Transformers for CLIP, and two interchangeable inference backends: a local MLX-VLM backend running a 4-bit quantised Qwen2.5-VL-7B on Apple Silicon, used during development, and a remote Qwen2.5-VL-7B-Instruct server exposed through an OpenAI-compatible API, used for the reported results. Panels are encoded as PNG and sent as base64 images. For the comparison to be fair, the baseline and our method are run on the same questions, in the same session, with the same prompt, the same frame budget C, and greedy decoding.

4 Results and Discussion

4.1 Experimental setup

Benchmarks. Video-MME [9] is a manually annotated multiple-choice benchmark of 2700 questions over 900 videos in six domains. We evaluate the medium (average 9 minutes) and long (average 41 minutes) splits, 1800 questions in total, without subtitles, since short videos are the setting where paneling is known to make little difference. MF2 [10] is a long-movie benchmark

in which each item is a pair of claims about a full-length movie, one true and one false, and the model judges each claim independently. We report both claim accuracy and *pair accuracy*, where a pair counts as correct only if both of its claims are judged correctly. At the time of writing, 455 of the 868 pairs have been evaluated.

Model and budget. All results use Qwen2.5-VL-7B-Instruct with $C = 32$ images per question, which corresponds to $4C = 128$ video frames for both methods, since both compose 2×2 panels. The baseline is our reimplementation of Video Panels with uniform sampling, equal-size cells, and $\gamma_{\text{fps}} = 1$.

4.2 Video-MME results

Table 2 shows overall accuracy. Our method improves on the paneling baseline by 0.9 points overall, 1.1 on medium videos and 0.6 on long videos.

Table 2: Video-MME accuracy (%) without subtitles, Qwen2.5-VL-7B-Instruct, $C = 32$. Higher is better.

Split	**Video Panels (baseline)**	**Ours**	**Δ**
Medium (n=900)	62.2	63.3	+1.1
Long (n=900)	53.4	54.0	+0.6
Overall (n=1800)	57.8	**58.7**	+0.9

A single accuracy figure hides how the two systems actually differ, so we also compare them question by question. The two methods agree on 1513 of the 1800 questions (84.1%). Among the 287 disagreements, our method is right on 103 and the baseline on 88; both are wrong on 656 questions and both right on 953. An exact McNemar test on the discordant pairs gives $p = 0.31$, so **the overall Video-MME improvement is not statistically significant at this sample size**. We consider it important to state this plainly: the direction is encouraging and consistent across both splits, but on this benchmark the evidence so far is that the two methods are close, not that ours is better.

The per-task breakdown in Table 3 is more informative than the overall number, because it shows a pattern rather than noise.

Table 3: Video-MME accuracy (%) by task category, sorted by improvement.

Task category	**n**	**Baseline**	**Ours**	**Δ**
Spatial Reasoning	29	72.4	79.3	+6.9
Attribute Perception	100	69.0	73.0	+4.0
Temporal Reasoning	164	40.2	43.9	+3.7
OCR Problems	82	57.3	61.0	+3.7
Object Reasoning	374	54.8	58.0	+3.2
Temporal Perception	37	59.5	62.2	+2.7
Action Recognition	182	55.5	57.1	+1.6
Counting Problem	143	39.9	40.6	+0.7
Spatial Perception	24	45.8	45.8	+0.0
Information Synopsis	241	76.3	74.7	−1.7
Action Reasoning	238	58.4	55.5	−2.9
Object Recognition	186	64.0	60.8	−3.2

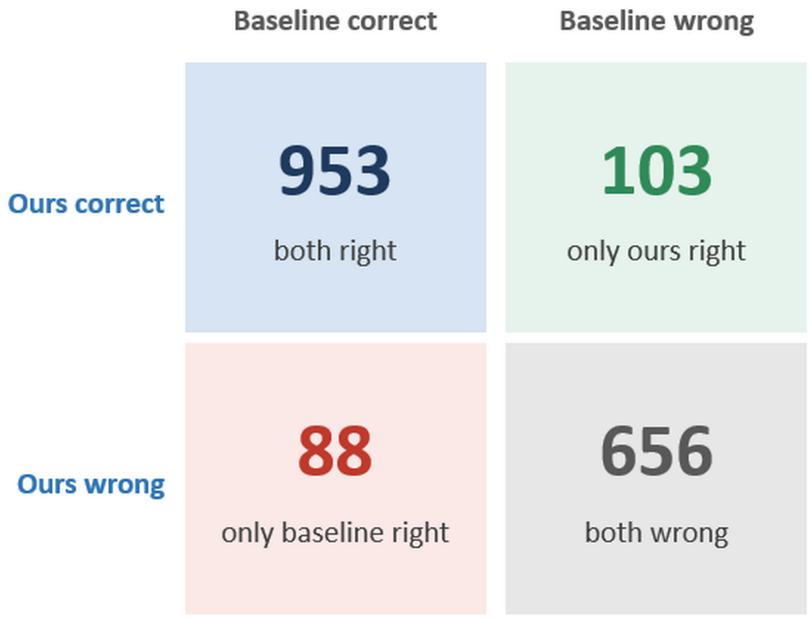


Figure 2: Question-level agreement between the baseline and our method on the same 1800 Video-MME questions.

The categories that improve are the ones the method was designed for. Spatial reasoning, attribute perception and OCR all depend on seeing something small clearly, which is exactly what enlarging a high-scoring cell provides. Temporal reasoning improves as well, which we attribute to windowed selection: the baseline's uniform sampling can miss a transition entirely, whereas our selector keeps the best-matching frame from every window of the timeline.

The categories that regress are equally coherent, and they point at a real cost of the method. Object recognition and action reasoning lose ground, and information synopsis, the highest-scoring category for both systems, loses 1.7 points. These are questions about the video as a whole rather than about a specific moment. For them, shrinking three cells to enlarge one is a bad trade: the answer may lie in any of the four, and an even, predictable layout serves the model better. Put differently, our method concentrates the pixel budget on the assumption that CLIP has correctly identified where the answer is. When that assumption holds the method wins, and when the question has no single locus, the concentration costs accuracy. The same pattern appears in the disagreement counts: our method's wins are concentrated in object reasoning (23) and temporal reasoning (15), while the baseline's wins are concentrated in object recognition (16) and action reasoning (16).

4.3 Question-level agreement

Because both systems answer the same 1800 questions, their outputs can be cross-tabulated directly. Table 4 and Figure 2 show the result. Identical answers are given on 1513 questions (84.1%), so the two methods differ on 287 questions, roughly one answer in six. Of those, 103 are questions our method gets right and the baseline does not, 88 are the reverse, and the remaining 96 are questions both answer incorrectly but with different letters. The 191 correctness flips nearly cancel, which is what produces the net effect of +15 questions behind the 0.9-point headline.

Two further observations follow from this table. First, the split of 103 against 88 on the 191 questions where correctness differs is close to what a coin flip would produce, which is the same conclusion the McNemar test reaches from the other direction. Second, there is a ceiling that neither method comes near breaking: 656 questions, 36% of the benchmark, are

Table 4: Question-level agreement on Video-MME (1800 questions).

	Baseline correct	**Baseline wrong**
Ours correct	953 (both right)	103 (only ours right)
Ours wrong	88 (only baseline right)	656 (both wrong)

answered incorrectly by both. Changing how the same 128 frames are presented cannot help when the evidence the question needs is not among those frames at all. Closing that gap requires a different frame budget or a stronger backbone, not a better layout.

4.4 Net questions gained by task

Accuracy percentages compress categories of very different sizes onto one scale, so a +6.9 on a category with 29 questions and a −3.2 on one with 186 look comparable when they are not. Converting each accuracy delta in Table 3 back into a count of questions removes that distortion. Table 5 and Figure 3 give the result: nine categories gain a total of +32 questions, three lose a total of −17, and the two sum to the net +15.

Table 5: Net questions gained or lost per task category on Video-MME (accuracy delta $\times n$). The column sums to $+32 - 17 = +15$.

Task category	**n**	**Net questions**
Object Reasoning	374	+12
Temporal Reasoning	164	+6
Attribute Perception	100	+4
OCR Problems	82	+3
Action Recognition	182	+3
Spatial Reasoning	29	+2
Counting Problem	143	+1
Temporal Perception	37	+1
Spatial Perception	24	0
Information Synopsis	241	−4
Object Recognition	186	−6
Action Reasoning	238	−7
Total	**1800**	**+15**

Read this way, the most reliable signals are the ones supported by large categories: Object Reasoning (+12) and Temporal Reasoning (+6) among the gains, and Action Reasoning (−7) and Object Recognition (−6) among the losses, all four with more than 160 questions. The categories with the largest percentage swings, such as Spatial Reasoning at +6.9 points, contribute only two questions each and should not be over-interpreted.

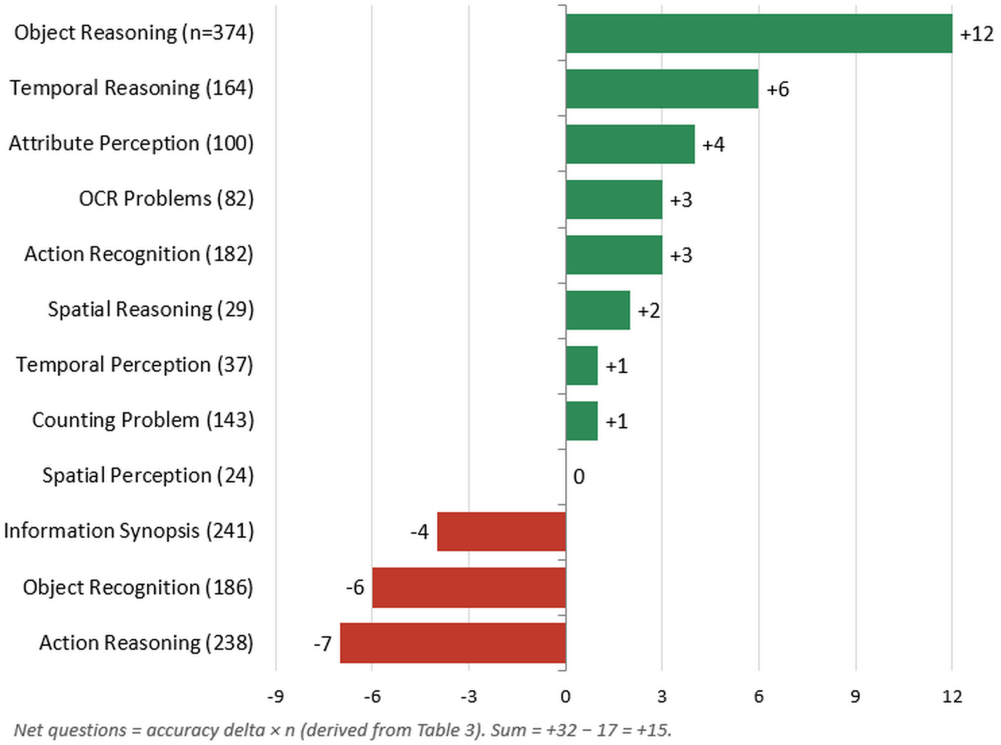


Figure 3: Net questions gained or lost by task category. Nine categories gain +32 questions in total and three lose −17.

4.5 Source of the disagreements

Figure 4 places the 191 correctness flips on the task axis, separating the questions won by our method from those won by the baseline. The two groups have distinct characters.

Where our method wins, the evidence is localised. CLIP finds the moment the question is about, that frame receives the largest cell in its panel, and objects and relations inside it become readable at a resolution the baseline never gives them. Windowed selection contributes independently by keeping one frame from every part of the timeline, so transitions that uniform sampling steps over are not skipped.

Where the baseline wins, the evidence is diffuse. Recognition and action questions can often be answered from any of the four cells in a panel. Shrinking three cells to enlarge one then removes evidence the model would otherwise have used, and an even, predictable layout serves these questions better. The net view in Table 5 shows the same contrast in aggregate: Object Reasoning gains +12 questions while Object Recognition loses −6.

4.6 MF2 results

MF2 is a harder and more diagnostic benchmark, since a model cannot succeed by guessing: answering every claim "false" yields 50% claim accuracy but 0% pair accuracy. Table 6 reports both, together with the model's answer distribution.

Pair accuracy improves from 16.9% to 23.5%, a relative gain of 39%. A pair-level McNemar test gives $p = 0.0016$: on 455 pairs, our method is right where the baseline is wrong 58 times and wrong where the baseline is right 28 times. This is the strongest evidence we have so far that the method does something real.

The mechanism is visible in the last three rows of the table. Both systems are heavily biased toward answering "false", which is what a model does when it cannot find evidence for a claim.

Table 6: MF2 results on 455 claim pairs (910 claims, 27 movies), video-only setting, Qwen2.5-VL-7B-Instruct.

Metric	**Baseline**	**Ours**	**Δ**
Claim accuracy	55.4	57.3	+1.9
Pair accuracy	16.9	**23.5**	+6.6
Accuracy on true claims	26.1	42.4	+16.3
Accuracy on false claims	84.8	72.1	−12.7
Proportion of "true" answers	20.7	35.2	+14.5

The baseline answers "true" for only 20.7% of claims, and our method for 35.2%. Our method gains 16 points on true claims and gives up 13 on false claims, which is a favourable trade under pair accuracy, and it suggests that query-aware selection is surfacing frames that actually confirm a claim rather than leaving the model to default to denial. Claim accuracy, by contrast, differs by only 1.9 points, which is why we treat pair accuracy as the metric of record on this benchmark.

The breakdown by claim granularity supports the same reading as Video-MME. Our method gains on Multi-scene claims (58.7 vs. 55.9) and Single-scene claims (54.9 vs. 52.5), and loses on Global claims (64.1 vs. 71.9), which concern the movie as a whole. Frame selection cannot help a question that is about everything, and uneven space allocation can hurt it.

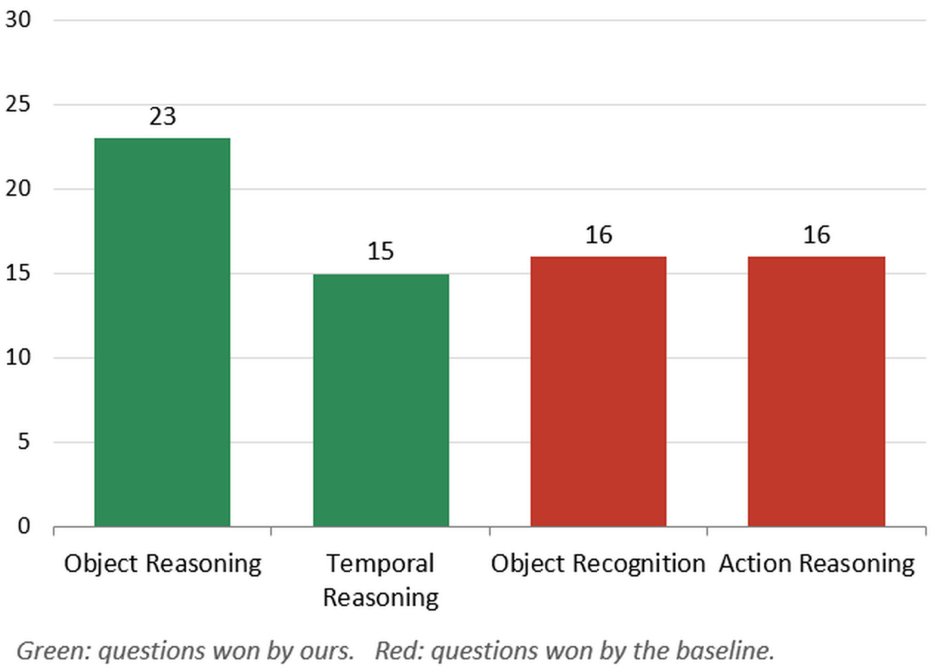


Figure 4: Where the two methods disagree, by task category. Questions won by our method are shown against questions won by the baseline.

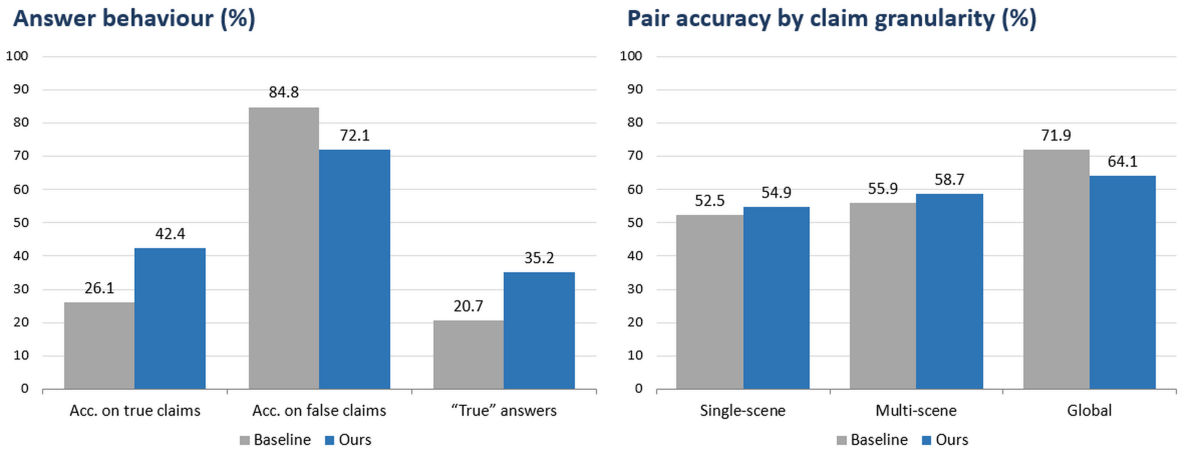


Figure 5: Left: how often each method answers "true", against the 50% rate of a balanced answerer. Right: pair accuracy by claim granularity. Surfacing more confirming evidence makes the model default to "false" less often, while Global claims, which concern the whole movie, do not benefit from frame selection.

Figure 5 shows both effects side by side. The answer-behaviour panel makes the size of the bias visible: MF2 is balanced by construction, so a system that judged claims on evidence alone would answer "true" about half the time, and both methods fall well short of that, the baseline more than ours. The granularity panel shows the limit of the approach, with the gains on Single- and Multi-scene claims set against the loss on Global claims.

4.7 Qualitative analysis

Figures 6 and 7 show one case of each kind. In the success case, the question asks about a detail that appears briefly; CLIP scores the frame containing it far above its neighbours, that frame receives the largest cell in its panel, and the model reads the detail correctly while the baseline, which shrinks the same frame to a quarter of the canvas, does not. In the failure case, the baseline answers correctly and our method does not: the relevance scores within the panel are close to uniform or point at the wrong moment, so the enlarged cell does not contain the evidence, and the three shrunken cells no longer show enough of it.

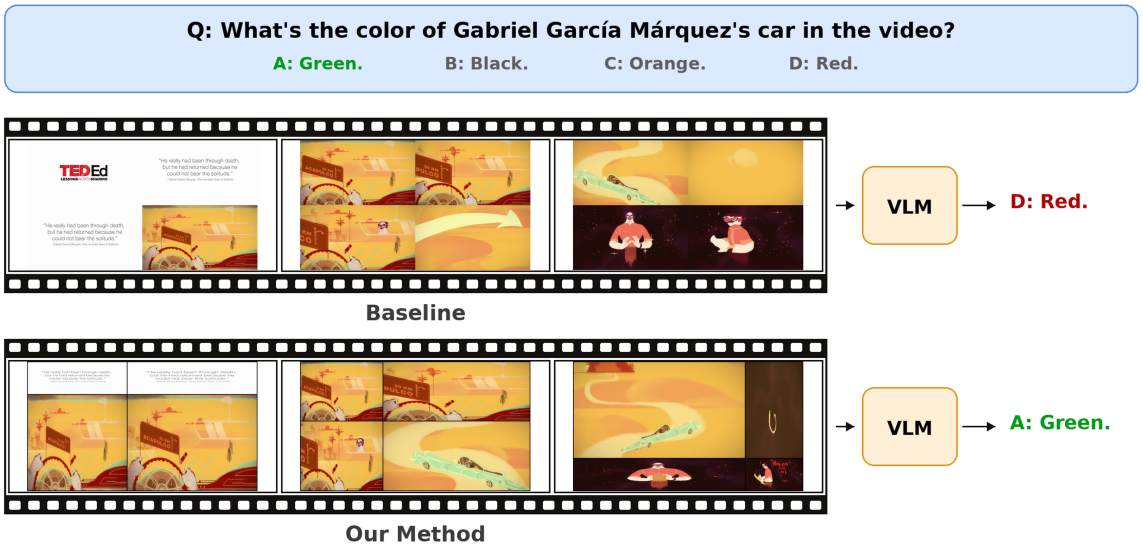


Figure 6: A case where our method answers correctly and the baseline does not.

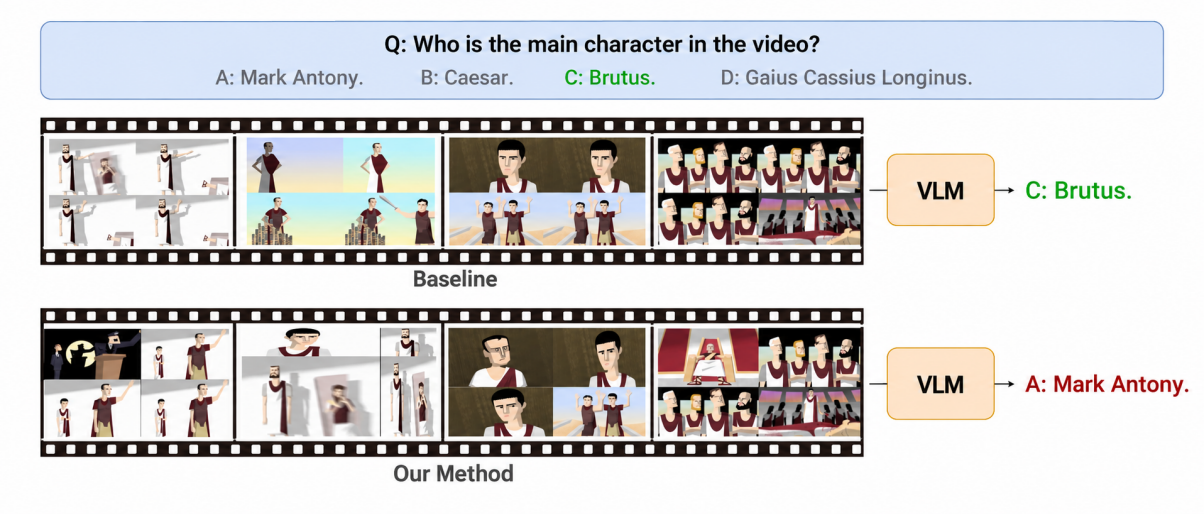


Figure 7: A case where the baseline answers correctly and our method does not.

4.8 Computational cost

The method is training-free, but it is not free. On our evaluation setup the baseline processes roughly one question every 16.5 seconds, while our method takes about 50 seconds, roughly three times slower. The difference is almost entirely the CLIP stage: the baseline decodes $4C$ frames and stops, whereas we decode up to $T_{\max}$ candidates and embed all of them. Because the scoring is query-dependent, the cost is paid once per question rather than once per video, which is the worst case for benchmarks like MF2 where one movie carries about sixteen questions. For MF2 we therefore cache each movie's candidate embeddings once and re-score per claim with a single text embedding and a dot product, which reduces the per-claim cost to near that of the baseline. The same caching applies to any deployment where several questions are asked about one video.

Figure 8 shows the comparison. The cost is paid per question rather than per video because the scoring depends on the query, which makes it most visible on benchmarks such as MF2, where roughly sixteen questions share a single movie; the embedding cache is what brings that case back down to baseline speed.

4.9 Limitations

Three limitations are worth stating clearly.

The layout cannot represent every score pattern. As shown in Section 3.5, a 2×2 cross-split has two degrees of freedom and four areas to fill, so anti-diagonal score patterns collapse to a near-uniform grid. In those panels the method silently degenerates into the baseline.

The method inherits CLIP's blind spots. CLIP scores a single frame against a text query, with no notion of motion, event order or dialogue. Where the answer depends on what was said, or on the order in which two events happened, the scores carry little signal and the resulting asymmetry is close to arbitrary. This is consistent with our weak results on Global MF2 claims and on information synopsis.

Run-to-run variation is not negligible. Serving a model with batched inference is not bit-deterministic, and on questions where the top two options are nearly tied, small numerical differences can flip the answer. At our sample sizes this is of the same order as some of the differences we are trying to measure, which is another reason to treat the 0.9-point Video-MME gap as provisional.

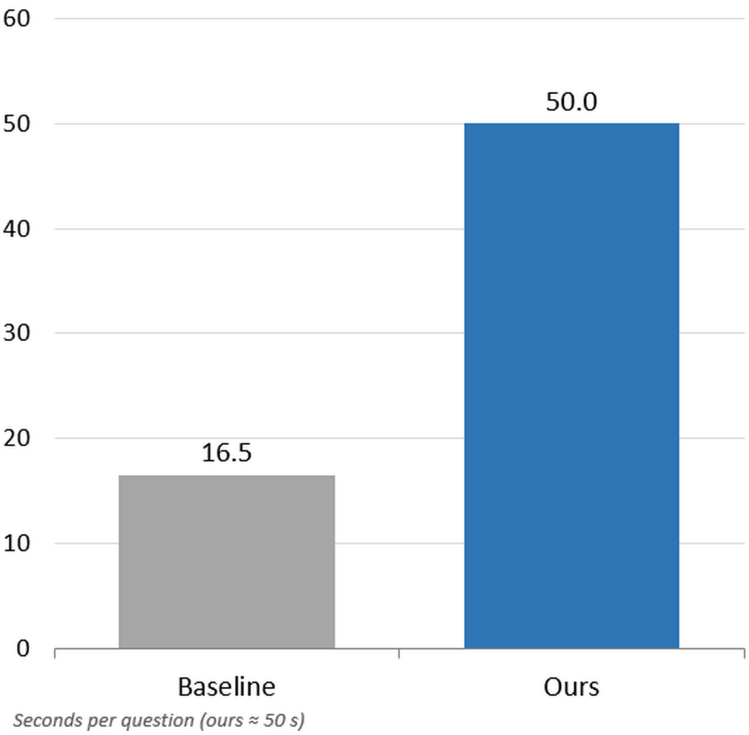


Figure 8: Seconds per question for the two methods. The gap is almost entirely the CLIP scoring stage, and it is removed by caching candidate embeddings when several questions share one video.

4.10 Current status and work plan

The experiments reported here are a first complete pass, not the final study. The following work remains.

1. **Finish MF2.** 455 of 868 pairs are done. Completing the benchmark will roughly double the sample and tighten the confidence in the pair accuracy result.
2. **Repeat runs for error bars.** Each configuration will be run several times so that accuracy can be reported with a spread rather than as a single number, and so that differences smaller than the run-to-run variation are not over-interpreted.
3. **Ablations.** Four components deserve isolation: windowed versus top-k selection; softmax temperature γ (how much asymmetry is useful); the relevance threshold τ_{raw}; and asymmetric versus symmetric composition with selection held fixed. The last is the critical one, since it separates the contribution of query-aware selection from that of adaptive sizing.
4. **A layout that escapes the cross-split constraint.** A treemap or slice-and-dice layout can realise arbitrary area targets. The open question is whether the resulting loss of a predictable reading order costs more than the gain in area fidelity.
5. **A stronger relevance signal.** Replacing CLIP with a longer-context or video-aware scorer, or adding a lightweight temporal smoothing term, should help precisely on the categories where we currently regress.
6. **Extension to multi-agent reasoning.** Our longer-term plan is to use the composed panels as the perceptual front-end of a lightweight multi-agent loop, in which a planner decides which temporal region deserves a second, higher-resolution pass. This targets the

questions that no single-pass method handles well, while keeping the number of model calls far below fully agentic systems such as Symphony.

4.11 Conclusion

We set out to test whether a VLM does better when its fixed pixel budget is spent unevenly, according to how relevant each frame is to the question. The evidence so far is mixed but instructive. On Video-MME the method is ahead by 0.9 points overall, which is not statistically significant, but the gains are concentrated exactly where the design predicts, in spatial, OCR and temporal questions, and the losses are concentrated in whole-video summary questions where the design's core assumption does not hold. On MF2 the improvement in pair accuracy is both larger and statistically significant, and it comes from a mechanism we can point to: the model finds more confirming evidence and defaults to "false" less often. Taken together, these results suggest that adaptive spatial allocation is worth pursuing, but that it should be conditioned on the kind of question being asked rather than applied uniformly, which is the direction our remaining work will take.

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